

REST API Reference: Hyperledger Fabric REC Service

This document provides request and response schemas for all Hyperledger Fabric REC endpoints.

All endpoints are available at `/api/ ...` and `/api/v1/ ...`.

1. Summary of Required Endpoints

Method	Path	Description
POST	<code>/api/rec</code>	Issue a new REC on Hyperledger Fabric via IssuerOrgMSP.
POST	<code>/api/rec/{recId}/transfer</code>	Transfer active REC units to another owner account.
POST	<code>/api/rec/{recId}/retire</code>	Permanently retire REC units for Scope 2 compliance.
POST	<code>/api/rec/{recId}/cancel</code>	Revoke / cancel a REC (RegulatorOrg / IssuerOrg).
GET	<code>/api/rec/{recId}</code>	Retrieve current on-chain state and off-chain metadata.
GET	<code>/api/rec/{recId}/history</code>	Retrieve complete chronological block transaction history.
GET	<code>/api/rec/{recId}/verify</code>	Cryptographically verify REC state and balance consistency.
POST	<code>/api/documents/upload</code>	Upload proof document, calculate SHA-256 fingerprint.
POST	<code>/api/rec/{recId}/verify-document</code>	Verify re-uploaded document against on-chain SHA-256 hash.
GET	<code>/api/fraud/alerts</code>	Retrieve active fraud alerts and double-counting anomalies.
GET	<code>/api/rec</code>	List all registered RECs with live ledger state.

2. Detailed Endpoint Specifications

2.1 Issue REC: `POST /api/rec`

- Request Body:

```
{
  "recId": "REC-000001",
  "generatorId": "GEN-SOLAR-01",
  "energySource": "SOLAR",
  "generationDate": "2026-09-10",
  "generationMWh": 100.0,
  "issuedQuantity": 100,
  "documentHash": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
}
```

- Success Response (`201 Created`):

```
{
  "message": "REC successfully issued on Hyperledger Fabric ledger.",
  "asset": {
    "recId": "REC-000001",
    "generatorId": "GEN-SOLAR-01",
    "energySource": "SOLAR",
    "generationDate": "2026-09-10",
    "generationMwh": 100,
    "issuedQuantity": 100,
    "currentOwner": "ISSUER-ORG",
    "activeQuantity": 100,
    "retiredQuantity": 0,
    "status": "ACTIVE",
    "documentHash": "e3b0c442 ... ",
    "createdAt": "2026-09-12T19:27:25.548Z",
    "updatedAt": "2026-09-12T19:27:25.548Z",
    "balances": {
      "ISSUER-ORG": 100
    }
  },
  "fraudEvaluation": {
    "recId": "REC-000001",
    "riskScore": 0.0,
    "riskLevel": "LOW",
    "reasons": []
  }
}
```

2.2 Transfer REC: `POST /api/rec/{recId}/transfer`

- **Request Body:**

```
{
  "fromOwner": "ISSUER-ORG",
  "toOwner": "CORP-BUYER-A",
  "quantity": 40,
  "transactionReference": "TRADE-REF-001",
  "callerOrg": "issuer"
}
```

- **Success Response (`200 OK`):**

```
{
  "message": "Successfully transferred 40 REC units to CORP-BUYER-A.",
  "asset": {
    "recId": "REC-000001",
    "currentOwner": "CORP-BUYER-A",
    "activeQuantity": 100,
    "retiredQuantity": 0,
    "balances": {
      "ISSUER-ORG": 60,
      "CORP-BUYER-A": 40
    }
  }
}
```

2.3 Retire REC: `POST /api/rec/{recId}/retire`

- Request Body:

```
{
  "owner": "CORP-BUYER-A",
  "quantity": 20,
  "retirementReason": "Scope 2 ESG Compliance Goal 2026",
  "callerOrg": "buyer"
}
```

- Success Response (`200 OK`):

```
{
  "message": "Successfully retired 20 REC units for owner CORP-BUYER-A.",
  "asset": {
    "recId": "REC-000001",
    "activeQuantity": 80,
    "retiredQuantity": 20,
    "balances": {
      "ISSUER-ORG": 60,
      "CORP-BUYER-A": 20
    }
  }
}
```

2.4 Cancel REC: `POST /api/rec/{recId}/cancel`

- Request Body:

```
{
  "reason": "Regulatory Directive: Fraudulent Telemetry",
  "callerOrg": "regulator"
}
```

- Success Response (`200 OK`):

```
{
  "message": "REC REC-000001 has been CANCELLED on the distributed ledger.",
  "asset": {
    "recId": "REC-000001",
    "status": "CANCELLED"
  }
}
```

2.5 Blockchain History: `GET /api/rec/{recId}/history`

- Success Response (`200 OK`):

```
{
  "recId": "REC-000001",
  "totalEvents": 3,
  "history": [
    {
      "txId": "2459b9f07daeb0f88 ... ",
      "timestamp": "2026-09-12T19:27:25.548Z",
      "isDelete": false,
      "value": {
        "recId": "REC-000001",
        "activeQuantity": 100,
        "retiredQuantity": 0,
        "currentOwner": "ISSUER-ORG"
      }
    },
    {
      "txId": "f908e23bb410ac901 ... ",
      "timestamp": "2026-09-12T19:27:28.120Z",
      "isDelete": false,
      "value": {
        "recId": "REC-000001",
        "activeQuantity": 100,
        "retiredQuantity": 0,
        "currentOwner": "CORP-BUYER-A"
      }
    },
    {
      "txId": "c345100ba9811ae23 ... ",
      "timestamp": "2026-09-12T19:27:31.125Z",
      "isDelete": false,
      "value": {
        "recId": "REC-000001",
        "activeQuantity": 80,
        "retiredQuantity": 20,
        "currentOwner": "CORP-BUYER-A"
      }
    }
  ]
}
```

2.6 Document Verification: `POST /api/rec/{recId}/verify-document`

- **Multipart Form Input:** `file: <binary_pdf>`
- **Matching Response (200 OK):**

```
{
  "match": true,
  "status": "DOCUMENT VERIFIED",
  "message": "Document byte integrity matches the immutable on-chain SHA-256 hash registered with this",
  "computed_hash": "e3b0c442 ... ",
  "registered_hash": "e3b0c442 ... ",
  "disclaimer": "Cryptographic match verifies byte-level integrity against the registered hash; legal",
}
```

- **Tampered Response (200 OK):**

```
{
  "match": false,
  "status": "DOCUMENT INTEGRITY FAILURE",
  "message": "Document SHA-256 hash does NOT match the registered blockchain hash! File has been altered",
  "computed_hash": "88a1b2c3 ... ",
  "registered_hash": "e3b0c442 ... ",
  "disclaimer": "Integrity check failed: bytes differ from the fingerprint committed to the Hyperledger Fabric ledger"
}
```